The examination is being carried out on the **following application documents**

Description, Pages

1-62 as published

Claims, Numbers

1-15 filed in electronic form on 31-01-2024

Drawings, Sheets

1/5-5/5 as published

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 WO 2019/067092 A1 (UNIV JOHNS HOPKINS [US] ET AL.) 4 April 2019 (2019-04-04)

D2 WO 2017/004243 A1 (CARIS SCIENCE INC [US]) 5 January 2017 (2017-01-05)

D3 US 2016/069889 A1 (SPETZLER DAVID [US] ET AL) 10 March 2016 (2016-03-10)

D4 US 2016/238616 A1 (MASSON LINDI [ZA] ET AL) 18 August 2016 (2016-08-18)

D5 WO 2019/200410 A1 (FREENOME HOLDINGS INC [US]) 17 October 2019 (2019-10-17)

D6 Li Jingjing ET AL: International Journal of Biological Sciences, vol. 16, no. 13, 1 January 2020 (2020-01-01), pages 2323-2339, XP093110139

D7 Knuchel Sarah ET AL: Oncotarget, Advance Publications, 5 May 2015 (2015-05-05), XP093209377

D8 Korbecki Jan ET AL: International journal of molecular sciences, vol. 21, no. 10, 25 May 2020 (2020-05-25), page 3723, XP093209378

D9 GIULIO C SPAGNOLI ET AL:INTERNATIONAL JOURNAL OF CANCER, JOHN WILEY & SONS, INC, US, vol. 86, no. 2, 28 March 2000 (2000-03-28), pages 238-243, XP071279304

D10 Messa Caterina ET AL: ACTA ONCOLOGICA., vol. 37, no. 3, 1 January 1998 (1998-01-01), pages 285-289, XP093209381

D11 US 2019/300964 A1 (ZHANG KANG [US] ET AL) 3 October 2019 (2019-10-03)

D12 US 2012/143805 A1 (GOLD LARRY [US] ET AL) 7 June 2012 (2012-06-07)

1 Il the amendments filed with the entry into the regional phase before the EPO meet the requirements of Article 123(2) EPC.

2 An International Preliminary Report on Patentability / International Preliminary Examination Report has already been drawn up for the present application in accordance with the PCT. The amendments filed with entry into the regional phase do not overcome the non-unity objections laid down in the international phase. The deficiencies mentioned in international report give rise to objections under the corresponding provisions of the EPC. The same reasoning as for the subject-matter of originally filed claims (partially repeated and further detailed here below) applies *mutatis mutandis* to the subject-matter of the new set of claims.

3 **Non-unity**

This Authority considers that the application does not meet the requirements of unity of invention and that there are 40 inventions covered by the claims indicated as follows:

Invention 1: claims:1-15(partially):

Invention 1.1 Method for detecting a colon cell proliferative disorder in a subject comprising: measuring in the biological sample an amount of a protein from a predetermined protein panel comprising the at least 6 following proteins: EGF, FGF-2, FLT3L, Fractalkine, IL-la, IL-2.

Invention 1.2 Method for detecting a colon cell proliferative disorder in a subject comprising: measuring in the biological sample an amount of a protein from a predetermined protein panel comprising the at least 6 following proteins: FGF-2, FLT3L, Fractalkine, IL-la, IL-2, CA19-9.

Inventions 1.3-1.28 Method for detecting a colon cell proliferative disorder in a subject comprising: measuring in the biological sample an amount of a protein from a predetermined protein panel comprising the at least 6 proteins selected from the following panel: FLT3L, Fractalkine, IL-la, IL-2, CA19-9, CEA, , HGF, OPN, CA125, AFP, CA15-3, CEACAM, CYFRA21-1, CA125, Kallikrein 6, Mesothelin, HE4, MIF, angiopoietin-2, mesothelin, MMP-2, TFMP1-2, tenascin C, Apo AI, factor H, AGRP.

Invention 1.29 Method for detecting a colon cell proliferative disorder in a subject comprising: measuring in the biological sample an amount of a protein from a predetermined protein panel comprising the at least 6 following proteins: TFMP1-2, tenascin C, Apo AI, factor H, AGRP, fetuin A.

Invention 2: claims:1-15(partially) Method for detecting a colon cell proliferative disorder in a subject comprising:measuring in the biological sample an amount of a protein from a predetermined protein panel comprising the at least 6 following proteins: FGF-2, FLT3L, Fractalkine, IL-la, IL-2, IL-6.

Inventions 3-39. claims: 1-15(partially) Method for detecting a colon cell proliferative disorder in a subject comprising:measuring in the biological sample an amount of a protein from a predetermined protein panel comprising at least 6 of proteins selected from the group consisting of: FLT3L, Fractalkine, IL-la, IL-2, IL-6, IL-8, GROa, MIP-3a, Complement component C2, Complement component C9, Factor D, Factor I, MBL, MMP-2, GDF-15, Osteonectin, Periostin, ANGPTL4, FGF-21, FGF-23, HGF, Angiopoietin-2, BMP-9, IL-1RII, HGFR, IL-6ra, OPN, Tenascin-C, Thrombospondin-2, uPAR, CD44, Kallikrein-6, Mesothelin, EpCAM, Apo A1, AGP, A2MB, Fetuin A, HP, L-Selectin, Complement component Clq, Complement component C3, Complement component C3b, Factor B, Factor H, Properdin, AGRP, MMP-12, CYFRA21-1, HE4, total PSA, MIF, AFP, CA125, CA19-9, CA15-3 (MUC-1).

Invention 40. claims: 1-15(partially): Method for detecting a colon cell proliferative disorder in a subject comprising: measuring in the biological sample an amount of a protein from a predetermined protein panel comprising the at least 6 proteins selected from the group consisting of: total PSA, EGF, FGF-2, FLT3L, Fractalkine, IL-la.

The reasons for which the inventions are not so linked as to form a single general inventive concept, as required by Article 82 EPC, are as follows:

It is considered that there are 40 inventions covered by the claims. The reasons, for which the inventions are not so linked as to form a single general inventive concept, are as follows:

(i) The common matter linking together the independent claims 1, 10, 14 and 15 is that at least 6 protein biomarkers are part of a panel and can be measured to assess a colon cell proliferative disorder (such as, to detect colon cell proliferative disorder; to classify and distinguish a population of subjects having a colon cell proliferative disorder from subjects not having the colon cell proliferative disorder; and to determine a protein profile of a biological sample) .

(ii) This common matter does not comprise a single general inventive concept, based on same or corresponding special technical features, because D1 discloses already assessing colon cell proliferative disorder (such as cancer) by measuring the amount of 6 biomarkers from a panel of biomarkers.

Thus D1 recites in para [181]:

"In some embodiments of methods provided herein that include **detecting** in one or more samples obtained from a subject the presence of: 1) one **or more genetic biomarkers** in one or more (e.g., 1, 2, 3, 4, 5, **6**, ... or 16) of the following genes: NRAS, CTNNBI, PIK3CA, FBXW7, APC, EGFR, BRAF, CDKN2A, PTEN, FGFR2, HRAS, KRAS, AKTI, TP53, PPP2R1A, and/or GNAS, 2) one or more (e.g., 1, 2, 3, 4, 5, **6,** 7, or 8) of the following protein biomarkers: **CA19-9, CEA, HGF, OPN, CA125, AFP**, prolactin, TIMP-1, and/or **CA15-3**, and 3) the presence of aneuploidy, the **subject is determined as having (e.g., diagnosed to have) or is determined to be (e.g. diagnosed as being) at elevated risk of having or developing one of the following types of cancer:** liver cancer, ovarian cancer, esophageal cancer, stomach cancer, pancreatic cancer, **colorectal cancer**, lung cancer, and/or breast cancer."

D1 further recites in para. [344]: "In some embodiments, methods provided herein include detecting the **presence of one or more peptides** (e.g., one or more peptides that are distinct from the various protein biomarkers described herein as being useful in one or more methods). In some embodiments, a subject is determined to be at elevated **risk of having or developing cancer if the biological sample contains one or more peptides indicative of cancer.**

In some embodiments, a subject is determined as having cancer if the biological sample **contains one or more peptides indicative of cancer**. In some embodiments, a peptide is derived from a protein (e.g., the peptide includes an amino acid sequence present in a protein biomarker or a different protein). **Non-limiting examples of peptides indicative of cancer** include the following peptides and peptides derived from the following proteins: **CEACAM, CYFRA21-1, CA125**, ... **Kallikrein 6**, ... **Mesothelin**, ...**HE4**, ... **MIF**, ... **angiopoietin-2**, ... **CA 19-9, CEA, CA 15-3,** CA-50, CA 72-4, OVX1, **mesothelin**, sialyl TN, **MMP-2**, -3, -7, -9, VAP-1, **TFMP1-2, tenascin C**, **Apo AI**, **factor H**, **AGRP, fetuin A**... etc."

D1 also claims: "claim 9. A method for **identifying the presence of cancer** in a subject comprising: detecting in a first biological **sample** obtained from the subject the presence of one **or more genetic biomarkers** in one or more of the following genes: NRAS, CTNNB1, PIK3CA, FBXW7, APC, EGFR, BRAF, CDKN2A, PTEN, FGFR2, HRAS, KRAS, AKT1, TP53, PPP2R1 A, or GNAS; detecting a level of one **or more (also 6)** of the following **protein biomarkers** in a second biological sample obtained from the subject: **CA19-9, CEA, HGF, OPN, CA125**, prolactin, TFMP-1, or MPO; comparing the detected levels of the one or more protein biomarker to one or more reference levels of the protein biomarkers; and identifying the presence of cancer in the subject when the presence of one or more genetic biomarkers is detected, the detected levels of the one or more protein biomarkers are higher than the reference levels of the one or more protein biomarkers, or both.

claim 13. The method of claim 9, wherein the level of each of **CA19-9, CEA, HGF, OPN, CA125**, prolactin, TIMP-1, and MPO is detected.

claim 15. The method of claim 9, **wherein the cancer i**s liver cancer, ovary cancer, esophageal cancer, stomach cancer, pancreatic cancer, **colorectal cancer**, lung cancer, breast cancer, or prostate cancer."

D1 also clearly discloses in para. [376] to [378]:

"[376] In some embodiments, the omega **score** (e.g., the omega score for the top mutation) can be used as an explanatory variable in a logistic regression. In some embodiments, the logistic regression can include one or more other explanatory variables (e.g., concentrations of proteins). In some embodiments, the **protein biomarkers** can be selected based on Mann- Whitney-Wilcoxon test. In some embodiments, the selected protein biomarkers have higher median values in cancer samples than in normal samples. In some embodiments, a forward selection is used to select explanatory variables from all biomarkers (including genetic biomarkers and protein biomarkers).

[377] **Applying a mathematical model** to the data can **generate one or more classifiers**. The **classifiers are mathematical model** with appropriate parameters (e.g., β coefficient in regression model). These parameters can be determined by applying a **mathematical model to a training data** set, e.g., a data set that includes both control subjects and a group of subjects that have cancer.

[378] A **classifier** can be evaluated for its ability to properly characterize each subject in a dataset (e.g., a **training dataset or a validation dataset**) using methods known to a person of ordinary skill in the art. Various statistical criteria can be used, for example, area under the curve (AUC), percentage of correct predictions, sensitivity, and/or specificity. In some embodiments, the classifier is evaluated by cross validation, Leave One OUT Cross Validation (LOOCV), n-fold cross validation, and jackknife analysis. In some embodiments, each classifier is evaluated for its ability to properly characterize those subjects in a dataset not used to generate the classifier (a "test dataset")."

Also paragraph [0043] of D1 discloses "Figure 8 contains a graph showing i**dentification of cancer type** by **supervised machine learning for patients classified by CancerSEEK** as positive. Percentages correspond to the proportion of patients correctly classified by one of the two most likely types (sum of light and dark blue bars) or the most likely type (light blue bar). Predictions for all patients for all cancer types are provided in Table 6. Error bars represent 95% confidence intervals."

And finally D1 also discloses in paragraph [972]: "The Bioplex 200 platform (Biorad, Hercules CA) was used to determine the concentration of multiple target proteins in the plasma samples. Luminex bead based immunoassays (Millipore, Bil erica NY) were performed following the manufacturers protocols and concentrations were determined using 5 parameter log curve fits (using Bioplex Manager 6.0) with vendor provided standards and quality controls. The HCCBP 1 MAG-58K panel was used to detect FGF2, Osteopontin, sFas, **IL-8**/ CXCL8, Prolactin, **HE4, HGF, AFP, CA125, IL6, CA15-3**, TGFa, **CYFRA21-1**, **CEA, CA19-9** and Leptin. The HANG2MAG- 12K panel was used to detect PAR, sPECAM-1, TSP-2, sEGFR, AXL and sHER2/sEGFR2/sErbB2. The HCMBMAG-22K panel was used to detect DKK1, GDF15, Osteoprotegerin (OPG) and Neuron-specific enolase (NSE). The HCCBP4MAG- 58K panels was used to detect **Kallikrein-6, CD44,** Midkine and **Mesothelin**. The HAGP1MAG-12K panel was used to detect Follistatin, G-CSF, **Angiopoietin-2** and (...)"

The common matter is thus disclosed and is not special. The same and corresponding features are not special, hence the above identified single general concept is not novel, let alone inventive, **in light of D1** (Rule 44(1) EPC).

As also mentioned during the international phase, **D11** discloses a predetermined protein panel characteristic of a colon cell proliferative disorder (methods and kits for identifying a subject as having colon cancer, Para. [0003); values measured for biomarkers of a biomarker panel. Para. (0162); cg27536151 references mitogen-activated protein kinase kinase 5 (MAP2K5). In some embodiments,cg1342UI12 references zinc finger protein 512B (ZNF512B). In some embodiments, cg14642259 references myosin binding protein C, cardiac (MYBPC3). Para. [0082], profile comprises six or more biomarkers. Para. [0069]) comprising at least 6 proteins (cg27536l51 references mitogen-activated protein kinase kinase 5 (MAP2K5). In some embodiments. cg13420112 references zinc finger protein 512B(ZNF512B) In some embodiments, cg14642259 references myosin binding protein C, cardiac (MYBPC3), Para. (0082); profile comprises six or more biomarkers, Para. (0069)); a classifier configured to distinguish a population of healthy subjects from subjects with a colon cell proliferative disorder (a binary classifier that distinguished two subsets of the total set of classes, Para. [0166), methods and kits for identifying a subject as having colon cancer. Para. [0003]; values measured for biomarkers of a biomarker panel. Para. [0162); multivariate model scores that distinguish between the presence or absence of colon cancer, between colon cancer subtypes, and between a "good" or a "poor*' prognosis, Para. [0170]) comprising: sets of measured values representative of proteins from a predetermined protein panel characteristic of the colon cell proliferative disorder (methods and kits for identifying a subject as having colon cancer. Para. [0003]; values measured for biomarkers of a biomarker panel. Para. (0162); cg27536151 references mitogen-activated protein kinase kinase 5 (MAP2K5).In some embodiments, cg13420112 references zinc finger protein 512B (ZNF512B). In some embodiments. cg14642259 references myosin binding protein C, cardiac (MYBPC3), Para. [0082]; profile comprises six or more biomarkers. Para. [0069]), wherein the sets of measured values are obtained from protein expression data from samples of healthy subject and samples of subjects having a colon cell proliferative disorder, wherein the measured values are used to generate a set of features corresponding to properties of the protein expression data (methods and kits for identifying a subject as having colon cancer, Para, [0003]; values measure for biomarkers of

a biomarker panel, Para. (0162); cg27536151 references mitogen-activated protein kinase kinase 5 (MAP2K5). In some embodiments.cg13420112 references zinc finger protein 512B (ZNF512B). In some embodiments, cg14642259 references myosin binding protein C. cardiac (MYBPC3). Para. [0082]; profile composes six or more biomarkers. Para. (0069); to generate a set of DNA fragments poor to submitting for sequencing analysis to generate CpG methylation data. In some cases, the sequencing analysis composes hybridizing each of the one or more biomarkers described herein with a probe, and performing a DNA sequencing reaction to quantify the methylation of each of the one or more biomarkers. In some Instances, the CpG methylation data Is then input into a machine learning/classification program to generate a methylation profile, Para. (0067)), wherein the sets of features is computer-processed using a machine learning or statistical model (a set of biological samples are generated and subsequently input into the machine learning/ classification program, Para. [0068]; better prediction is achieved by being more liberal about the biomarker panels used as features, Para. (0163)), and wherein the machine learning or statistical model provides a feature vector useful as a classifier capable of distinguishing a population of healthy subjects from subjects having the colon cell proliferative disorder (the machine learning method utilizes an algorithm selected from one or more of the following: a principal component analysis, a logistic regression analysis, a nearest neighbour analysis, a support vector machine, and a neural network model. Para. [0068]; difference dataset composes (i) a first difference between the methylation profile of the treated genomic DNA with a methylation profile of a first normal sample; (Ii) a second difference between a methylation profile of a second normal sample and a methylation profile of a third normal sample. Para (0077)). a system comprising a machine learning model classifier for detecting a colon cell proliferative disorder (a set of biological samples are generated and subsequently input into the machine learning/classification program, Para. (0068); multivariate model scores that distinguish between the presence or absence of colon cancer, between colon cancer subtypes, and between a "good" or a "poor'' prognosis, Para. (0170]) comprising: a computer-readable medium comprising a classifier operable to classify subjects based on a predetermined protein panel (a set of biological samples are generated and subsequently input into the machine learning/classification program. Para. (0068); multivariate model scores that distinguish between the presence or absence of colon cancer, between colon cancer subtypes, and between a "good" or a "poor" prognosis, Para. (0170)); a method for determining a protein profile of a biological sample from a subject (methods and kits for identifying a subject as having colon cancer, Para. (0003); values measured for biomarkers

of a biomarker panel, Para (0162); obtained from a biological sample from the subject suspected of having colon cancer. Para. (0004)) composing: a) obtaining the biological sample containing proteins from the subject (obtained from a biological sample from the subject suspected of having colon cancer, Para. [0004]); and b) measuring in the biological sample an amount of a protein from a predetermined panel of proteins comprising at least 6 proteins, thereby providing the protein profile of the subject (methods and kits for identifying a subject as having colon cancer, Para. (0003); values measured for biomarkers of a biomarker panel, Para. (0162); cg27536151 references mitogen-activated protein kinase kinase 5 (MAP2K5). In some embodiments.cg13420112 references zinc finger protein 512B (ZNF512B). In some embodiments, cg14642259 references myosin binding protein C. cardiac (MYBPC3), Para. (0082]; profile comprises six or more biomarkers, Para. [0069]); a method for detecting a colon cell proliferative disorder in a subject (methods and kits for identifying a subject as having colon cancer, Para. (0003), values measured for biomarkers of a biomarker panel. Para. (0162]; obtained from a biological sample from the subject suspected of having colon cancer, Para. [0004]).

Further D12 teaches one or more processors for executing instructions stored on the computer-readable medium (The computer program product includes a computer readable medium embodying program code executable by a processor of a computing device or system, Para. (0055)).

The common matter is thus disclosed and is not special. The same and corresponding features are not special, hence the above identified single general concept is not new, let alone inventive, **in light of D11 alone (or combined with D12)** (Rule 44(1) EPC).

(iii) The common matter linking the biomarkers panel and assessing colon cell proliferative disorder methods according to claims 1, 10, 14 and 15 does not involve any same or corresponding special technical features:

Each partial claims 1, 10, 14 and 15 relate to assessing a colon cell proliferative disorder by determining the level of at least 6 markers from a panel of biomarkers. Each special technical feature relates to at least one protein from a panel of at least 6 biomarkers selected from: EGF, FGF-2, FLT3L, Fractalkine, IL-la, IL-2, IL-6, IL-8, GROa, MIP-3a, Complement component C2, Complement

component C9, Factor D, Factor I, MBL, MMP-2, GDF-15, Osteonectin, Periostin, ANGPTL4, FGF-21, FGF-23, HGF, Angiopoietin-2, BMP-9, IL-1RII, HGFR, IL-6ra, OPN, Tenascin-C, Thrombospondin-2, uPAR, CD44, Kallikrein-6, Mesothelin, EpCAM, Apo A1, AGP, A2MB, Fetuin A, HP, L-Selectin, Complement component Clq, Complement component C3, Complement component C3b, Factor B, Factor H, Properdin, AGRP, MMP-12, CYFRA21-1, HE4, total PSA, MIF, AFP, CA125, CA19-9, CA15-3 (MUC-1), and CEA respectively.

Hence, each partial claims 1, 10, 14 and 15 do not share same or corresponding features.

(iv) As each of the partial claims 1, 10, 14 and 15 comprise neither the same, nor corresponding special technical features, the technical relationship between the subject matter of the claims is lacking and the claims are not so linked as to form a single general inventive concept (Rule 44(2) EPC). The application does not fulfil the requirement for unity of invention (a combination of a posteriori and a priori lack of unity) (Article 82 EPC).

2 The search for subject 2-40 represent a major extra search burden. In consequence the applicant was invited to pay one additional search fees but he refused to do so. An opinion will be given on the searched subject-matter (invention 1.1-1.29) .

3 **Novelty**

3.1 Document D1 discloses (see references in the search report and here below):

3.1.1 A method for detecting a colon cell proliferative disorder in a subject comprising:
a) obtaining a biological sample containing proteins from the subject;
b) measuring in the biological sample an amount of a protein from a predetermined protein panel comprising at least 6 of proteins selected from the group consisting of: FGF2, IL-6, IL-8, CA19-9, CEA, , HGF, OPN, CA125, AFP, CA15-3, CEACAM, CYFRA21-1, CA125, Kallikrein 6, Mesothelin, HE4, MIF, angiopoietin-2, mesothelin, MMP-2, TFMP1-2, tenascin C, Apo AI, factor H, AGRP, TFMP1-2, tenascin C, Apo AI, factor H, AGRP and fetuin A thereby providing a protein profile of the subject; and
c) processing the protein profile using a machine learning model trained to be capable of distinguishing between subjects with the colon cell proliferative disorder and subjects without the colon cell proliferative disorder to provide an

output value associated with presence or absence of the colon cell proliferative disorder, thereby indicating the presence or absence of the colon cell proliferative disorder in the subject.

3.1.2 A predetermined protein panel characteristic of a colon cell proliferative disorder comprising at least 6 proteins selected from the group consisting of: FGF2, IL-6, IL-8, CA19-9, CEA, , HGF, OPN, CA125, AFP, CA15-3, CEACAM, CYFRA21-1, CA125, Kallikrein 6, Mesothelin, HE4, MIF, angiopoietin-2, mesothelin, MMP-2, TFMP1-2, tenascin C, Apo AI, factor H, AGRP, TFMP1-2, tenascin C, Apo AI, factor H, AGRP and fetuin A.

3.1.3 A classifier configured to distinguish a population of subjects having a colon cell proliferative disorder from subjects not having the colon cell proliferative disorder, comprising:
sets of measured values representative of proteins from a predetermined protein panel of six biomarkers selected from FGF2, IL-6, IL-8, CA19-9, CEA, , HGF, OPN, CA125, AFP, CA15-3, CEACAM, CYFRA21-1, CA125, Kallikrein 6, Mesothelin, HE4, MIF, angiopoietin-2, mesothelin, MMP-2, TFMP1-2, tenascin C, Apo AI, factor H, AGRP, TFMP1-2, tenascin C, Apo AI, factor H, AGRP and fetuin A, characteristic of the colon cell proliferative disorder,
wherein the sets of measured values are obtained from protein expression data from subjects having the colon cell proliferative disorder and subjects not having the colon cell proliferative disorder,
wherein the measured values are used to generate a set of features corresponding to properties of the protein expression data, wherein the set of features is computer-processed using a machine learning or statistical model, and
wherein the machine learning or statistical model provides a feature vector useful as a classifier capable of distinguishing the population of subjects having the colon cell proliferative disorder from subjects not having the colon cell proliferative disorder.

3.1.4 A method for determining a protein profile of a biological sample from a subject comprising:
a) obtaining the biological sample containing proteins from the subject; and
b) measuring in the biological sample an amount of a protein from a predetermined protein panel comprising at least 6 proteins selected from the group consisting of: FGF2, IL-6, IL-8, CA19-9, CEA, , HGF, OPN, CA125, AFP, CA15-3, CEACAM, CYFRA21-1, CA125, Kallikrein 6, Mesothelin, HE4, MIF,

angiopoietin-2, mesothelin, MMP-2, TFMP1-2, tenascin C, Apo AI, factor H, AGRP, TFMP1-2, tenascin C, Apo AI, factor H, AGRP and fetuin A, thereby providing the protein profile of the subject.

3.2 Thus D1 recites in para [181]:

"In some embodiments of methods provided herein that include **detecting** in one or more samples obtained from a subject the presence of: 1) one **or more genetic biomarkers** in one or more (e.g., 1, 2, 3, 4, 5, **6**, 7, 8, 9, 10, 11, 12, 13, 14, 15, or 16) of the following genes: NRAS, CTNNBI, PIK3CA, FBXW7, APC, EGFR, BRAF, CDKN2A, PTEN, FGFR2, HRAS, KRAS, AKTI, TP53, PPP2R1A, and/or GNAS, 2) one or more (e.g., 1, 2, 3, 4, 5, 6, 7, or 8) of the following protein biomarkers: **CA19-9, CEA, HGF, OPN, CA125, AFP**, prolactin, TIMP-1, and/or **CA15-3**, and 3) the presence of aneuploidy, the **subject is determined as having (e.g., diagnosed to have) or is determined to be (e.g. diagnosed as being) at elevated risk of having or developing one of the following types of cancer:** liver cancer, ovarian cancer, esophageal cancer, stomach cancer, pancreatic cancer, **colorectal cancer**, lung cancer, and/or breast cancer."

D1 further recites in para [198] "A peptide biomarker can be any appropriate peptide biomarker. In some embodiments, a peptide biomarker can be a peptide biomarker associated with cancer. For example, a peptide biomarker can be a peptide having elevated levels in a cancer (e.g., as compared to a reference level of the peptide). Examples of peptide biomarkers include, without limitation, AFP, Angiopoietin-2, AXL, CA125, CA15-3, CA19-9, CD44, CEA, CYFRA 21-1, DKK1, Endoglin, **FGF2**, Follistatin, Galectin-3, G-CSF, GDF15, HE4, HGF, IL-6, IL-8, Kallikrein-6, Leptin, LRG-1, Mesothelin, Midkine, Myeloperoxidase, NSE, OPG, OPN1.2 , PAR,..."

D1 further recites in para [344]: "In some embodiments, methods provided herein include detecting the **presence of one or more peptides** (e.g., one or more peptides that are distinct from the various protein biomarkers described herein as being useful in one or more methods). In some embodiments, a subject is determined to be at elevated **risk of having or developing cancer if the biological sample contains one or more peptides indicative of cancer.** In some embodiments, a subject is determined as having cancer if the biological sample **contains one or more peptides indicative of cancer**. In some embodiments, a peptide is derived from a protein (e.g., the peptide includes an amino acid sequence present in a protein biomarker or a different protein). **Non-limiting examples of peptides indicative of cancer** include the following

peptides and peptides derived from the following proteins: **CEACAM, CYFRA21-1, CA125**, ... **Kallikrein 6**, ... **Mesothelin**, ...**HE4**, ... **MIF**, ... **angiopoietin-2**, ... **CA 19-9, CEA, CA 15-3,** CA-50, CA 72-4, OVX1, **mesothelin**, sialyl TN, **MMP-2**, -3, -7, -9, VAP-1, **TFMP1-2, tenascin C**, **Apo AI**, **factor H**, **AGRP, fetuin A**... etc."

D1 also claims: "9. A method for **identifying the presence of cancer** in a subject comprising: detecting in a first biological sample obtained from the subject the presence of one or more genetic biomarkers in one or more of the following genes: NRAS, CTNNB1, PIK3CA, FBXW7, APC, EGFR, BRAF, CDKN2A, PTEN, FGFR2, HRAS, KRAS, AKT1, TP53, PPP2R1 A, or GNAS; detecting a level of one **or more (also 6)** of the following protein biomarkers in a second biological sample obtained from the subject: **CA19-9, CEA, HGF, OPN, CA125**, prolactin, TFMP-1, or MPO; comparing the detected levels of the one or more protein biomarker to one or more reference levels of the protein biomarkers; and identifying the presence of cancer in the subject when the presence of one or more genetic biomarkers is detected, the detected levels of the one or more protein biomarkers are higher than the reference levels of the one or more protein biomarkers, or both.

13. The method of claim 9, wherein the level of each of **CA19-9, CEA, HGF, OPN, CA125**, prolactin, TIMP-1, and MPO is detected.

15. The method of claim 9, **wherein the cancer i**s liver cancer, ovary cancer, esophageal cancer, stomach cancer, pancreatic cancer, **colorectal cancer**, lung cancer, breast cancer, or prostate cancer."

D1 also clearly discloses:

"[376] In some embodiments, the omega **score** (e.g., the omega score for the top mutation) can be used as an explanatory variable in a logistic regression. In some embodiments, the logistic regression can include one or more other explanatory variables (e.g., concentrations of proteins). In some embodiments, the **protein biomarkers** can be selected based on Mann- Whitney-Wilcoxon test. In some embodiments, the selected protein biomarkers have higher median values in cancer samples than in normal samples. In some embodiments, a forward selection is used to select explanatory variables from all biomarkers (including genetic biomarkers and protein biomarkers).

[377] **Applying a mathematical model** to the data can **generate one or more classifiers**. The **classifiers are mathematical model** with appropriate parameters (e.g., β coefficient in regression model). These parameters can be

determined by applying a **mathematical model to a training data** set, e.g., a data set that includes both control subjects and a group of subjects that have cancer.

[378] A **classifier** can be evaluated for its ability to properly characterize each subject in a dataset (e.g., a **training dataset or a validation dataset**) using methods known to a person of ordinary skill in the art. Various statistical criteria can be used, for example, area under the curve (AUC), percentage of correct predictions, sensitivity, and/or specificity. In some embodiments, the classifier is evaluated by cross validation, Leave One OUT Cross Validation (LOOCV), n-fold cross validation, and jackknife analysis. In some embodiments, each **classifier is evaluated for its ability to properly characterize those subjects in a dataset** not used to generate the classifier (a "test dataset")."

Also paragraph [43] of D1 discloses "Figure 8 contains a graph showing i**dentification of cancer type** by **supervised machine learning for patients classified by CancerSEEK** as positive. Percentages correspond to the proportion of patients correctly classified by one of the two most likely types (sum of light and dark blue bars) or the most likely type (light blue bar). Predictions for all patients for all cancer types are provided in Table 6. Error bars represent 95% confidence intervals."

And finally D1 also discloses in paragraph [972]: "The Bioplex 200 platform (Biorad, Hercules CA) was used to determine the concentration of multiple target proteins in the plasma samples. Luminex bead based immunoassays (Millipore, Bil erica NY) were performed following the manufacturers protocols and concentrations were determined using 5 parameter log curve fits (using Bioplex Manager 6.0) with vendor provided standards and quality controls. The HCCBP 1 MAG-58K panel was used to detect FGF2, Osteopontin, sFas, **IL-8**/ CXCL8, Prolactin, **HE4, HGF, AFP, CA125, IL6, CA15-3**, TGFa, **CYFRA21-1**, **CEA, CA19-9** and Leptin. The HANG2MAG- 12K panel was used to detect PAR, sPECAM-1, TSP-2, sEGFR, AXL and sHER2/sEGFR2/sErbB2. The HCMBMAG-22K panel was used to detect DKK1, GDF15, Osteoprotegerin (OPG) and Neuron-specific enolase (NSE). The HCCBP4MAG- 58K panels was used to detect **Kallikrein-6, CD44,** Midkine and **Mesothelin**. The HAGP1MAG-12K panel was used to detect Follistatin, G-CSF, **Angiopoietin-2** and (...)"

D1 also teaches in para [348] that: "Any genetic biomarker (e.g., genetic alteration) that is associated with cancer can be verified using methods described herein, such as **FGF2**...".

3.3 The subject-matter of claims 1, 10, 14 and 15 therefore differs from this known D1 in that in the present application the selected 6 biomarkers panel is EGF, FGF-2, FLT3L, Fractalkine, IL-la, IL-2.

3.4 Therefore, the subject-matter of claims 1, 10, 14 and 15 is new according to Article 54 EPC.

4 Inventive step

4.1 There is no technical effect deriving from the difference.

The problem to be solved is to provide an alternative of 6 biomarkers of colon cell proliferative disorders.

The solution is to select as the 6 biomarkers the following proteins: EGF, FGF-2, FLT3L, Fractalkine, IL-la, IL-2.

4.1.1 As demonstrated above most of the biomarkers of present application are already well known cancer and/or colon biomarkers. All cancer and colon biomarkers disclosed in the present application appear to be anticipated by the disclosure of D1, but also of D2 to D10.

4.1.2 For example, D2 discloses the following colorectal specific biomarkers (see table 4 pages 110-111):

Colorectal cancer CD9, EGFR, NGAL, CD81, STEAP, CD24, A33, CD66E, EPHA2, Ferritin, GPR30, GPRI IO, MMP9, **OPN**, p53, TMEM211, TROP2, TGM2, TIMP, EGFR, DR3, UNC93A, MUC17, **EpCAM**, MUCI, MUC2, TSG101, CD63, B7H3

Colorectal cancer A33, **AFP**, ... , **CA-19-9**, CCSA-2, CCSA-3&4, CD10, CD24, **CD44**, CD63, CD66 CEA, CD66e **CEA**, CD81, CD9, CD A, C-Erb2, CRMP-2, CRP, CRTN, CXCL12, **CYFRA21-1,** DcR3, DLL4, DR3, EGFR, **Epcam**, ... **MIF**, MMP7, MMP9, MS4A1, MUCI, MUC17, MUC2, Ncam, NGAL, NNMT, **OPN**, p53, PCSA, PDGFRB, PRL, PSMA, PSME3, Reg IV, SCRN1, Sept-9, SPARC, SPON2, SPR, SRVN, TFF3, TGM2, TIMP-1, TMEM211, TNF- alpha, TP A, TPS, Trail-R2, Trail-R4, TrKB, TROP2, Tsg 101, TWEAK, UNC93A, VEGFA (...) **CD44**, CD46, CD55, CD59, CD63, CD81, CD82, CD9, CDC42, CDH1, CDH17, **CEACAM5**,...

D2 further discloses: " [00491] The tumor marker can be a generic tumor marker or be associated with certain tumor types, such as those originating from different anatomical origins. In an embodiment, the tumor marker can be chosen to correspond to a certain tumor type. For example, exemplary **tumor markers** and associated tumor types include without limitation the following, listed as antigen (optional name) (cancer types): Alpha fetoprotein (AFP) (germ cell tumor, hepatocellular carcinoma); CA15-3 (breast cancer); CA27-29 (breast cancer); **CA19-9** (mainly pancreatic cancer, but also **colorectalcancer and other types of gastrointestinalcancer**); **CA-125** (ovarian cancer, endometrial cancer, fallopian tube cancer, lung cancer, breast cancer and **gastrointestinal cancer**) (...).

Additional **tumor types and associated biomarkers comprise the following, listed as tumor type (markers)**: **Colorectal** (M2-PK, **CEA, CA19-9, CA125**); Breast (CEA, CA15-3, Cyfra 21-1); Ovary (CEA, CA19-9, CA125, AFP, BHCG); Uterine (CEA, CA19-9, CA125, Cyfra 21-1, SCC); Prostate (PSA); Testicle (AFP, BHCG); Pancreas/Stomach (CEA, CA19-9, CA 72-4); Liver (CEA, AFP); Oesophagus (CEA, Cyfra 21-1); Thyroid (CEA, NSE); Lung (CEA, CA19-9, CA125, NSE, Cyfra 21-1); Bladder (CEA, Cyfra 21-1, TP A).

One or more of these markers can be used as the target biomarker recognized by the variable region of the multipartite construct of the invention."

D3 discloses as well the biomarker of present application, and the use of at least six of those biomarkers to detect/diagnose colon cancer.

Thus D3 recites in Table 3: "Colorectal cancer CD9, EGFR, NGAL, CD81, STEAP, CD24, A33, CD66E, EPHA2, Ferritin, GPR30, GPR110, MMP9, **OPN**, p53, TMEM211, TROP2, TGM2, TIMP, EGFR, DR3, UNC93A, MUC17, **EpCAM**, MUC1, MUC2, TSG101, CD63, B7H3 Colorectal cancer DR3, STEAP, epha2, TMEM211, unc93A, A33, CD24, NGAL, EpCam, MUC17, TROP2, TETS Colorectal cancer A33, **AFP**, ALIX, ALX4, ANCA, APC, ASCA, AURKA, AURKB, B7H3, BANK1, BCNP, BDNF, **CA-19-9**, CCSA-2, CCSA-3&4, CD10, CD24, **CD44**, CD63, CD66 CEA, CD66e, **CEA**, CD81, CD9, CDA, C-Erb2, CRMP-2, CRP, CRTN, CXCL12, **CYFRA21-1**, DcR3, DLL4, DR3, EGFR, **Epcam**, EphA2, FASL, FRT, GAL3, GDF15, GPCR (GPR110), GPR30, GRO-1, HBD 1, HBD2, HNP1-3, IL-1B, **IL8**, IMP3, L1CAM, LAMN, MACC-1, MGC20553, MCP-1, M-CSF, MIC1, **MIF**, MMP7, MMP9, MS4A1, MUC1, MUC17, MUC2, Ncam, NGAL, NNMT, OPN, p53, PCSA, PDGFRB, PRL, **PSMA**, PSME, **CEACAM5, CA125, IL-1 alpha,**(...) ."

D3 specifies in paragraph [0166]: "Any of the types of biomarkers described herein can be assessed using the subject methods and compositions. Exemplary biomarkers include without limitation those in **Table 3**. The markers can be detected **as protein** or as mRNA, which can be circulating freely or in a complex with other biological molecules. As appropriate, the markers in **Table 3** can also be used for capture and/or detection of vesicles for characterizing phenotypes as disclosed herein. In some cases, multiple capture and/or detectors are used to enhance the characterization."

In paragraphs 64 and 65 of D3 is specified that: "A **biosignature** can be used in any methods disclosed herein, e.g., to assess whether a subject is afflicted with disease, is at risk for developing disease or to assess the stage or progression of the disease. For example, a biosignature can be used to assess whether a subject has prostate cancer, **colon cance**r, or other cancer as described herein. Furthermore, **a biosignature can be used to determine a stage of a disease or condition, such as colon cancer**. The biosignature/**biomarker profile** comprising a microvesicle can include assessment of payload within the microvesicle."

Furthermore:

D6 discloses the biomarkers IL-la, IL-2 of colon cell proliferative disorders (see the entire document). D7 discloses the biomarker FGF-2 of colon cell proliferative disorders (see the entire document). D8 discloses the biomarker Fractalkine of colon cell proliferative disorders (see the entire document). D9 discloses the biomarker FLT3L of colon cell proliferative disorders (see the entire document). D10 discloses the biomarker EGF of colon cell proliferative disorders (see the entire document).

No single biochemical marker in circulation will ever meet the sensitivity and specificity criteria required for disease screening. Therefore, it is a common practice to use a marker panel to predict the presence or absence of a disease.

The person skilled in the art knows from D1-D10 (see passages cited above) that EGF, FGF-2, FLT3L, Fractalkine, IL-la, IL-2 are all markers of colon cell proliferative disorders (such as colorectal cancer). The common practice of combining biochemical markers will consequently prompt him to use one or more of these markers together in order to assess colon cell proliferative disorders. Furthermore, panel of 6 biomarkers to assess colon cell proliferative disorders are also known from any of D1 to D3.

The subject-matter of claims 1, 10, 14 and 15 consists in the selection of the 6 biomarkers: EGF, FGF-2, FLT3L, Fractalkine, IL-Ia, IL-2 from the biomarkers known in the art. Such a selection can only be regarded as inventive if the combination of these 6 biomarkers presents unexpected effects or properties in relation to the rest of the range. However, no such effects or properties are indicated in the application. Hence, no inventive step is present in the subject-matter of claims 1, 10, 14 and 15 (Articles 52(1) and 56 EPC).

4.1.3 Dependent claims2-9, 11-13 do not contain any features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC in respect of novelty and/or inventive step (Article 54 and 56 EPC) for the following reasons: The features of dependent claims 2-9, 11-13 are either known from D1-D10 (see citations above) or fall within the knowledge and the ability of a person skilled in the art.

5 Clarity

The application does not meet the requirements of Article 84 EPC, because independent claims 1, 10, 14 and 15 are not clear.

The subject-matter of the independent claims 1, 10 and 15 as drafted could be interpreted as referring only to measuring one biomarker and not 6 biomarkers together. Thus, the technical feature "measuring in the biological sample an amount of **a protein** from" a panel, could be interpreted as meaning measuring the amount of only one biomarker of a panel, irespective of the size of the panel.

Claim 14 is not limited to the panel of 6 disclosed in claims 1, 10 and 15. It was interpreted that subject-matter of claim 14 was as well limited to the biomarker of claim 1. In absence of such a limitation the subject-matter of claim 14 is not new in view of the teaching of D1 (see references and citations above).

6 Conclusion

6.1 The applicant is requested to file new claims which take account of the above comments.

6.2 In order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant is requested to clearly identify the amendments carried out, irrespective of

whether they concern amendments by addition, replacement or deletion, and to indicate the passages of the application as filed on which these amendments are based.

6.3 If the applicant regards it as appropriate these indications could be submitted in handwritten form on a copy of the relevant parts of the application as filed.